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RESEARCH ARTICLE

Biochar field testing in the UK: outcomes and implications for use



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Jim Hammond*[1], Simon Shackley[1], Miranda Prendergast-Miller[1], Jason Cook[1], Sarah Buckingham[2] & Valentini A Pappa[2]

Background: There is a lack of biochar field trials in temperate climate regions. Wood biochar was applied during 2009–2011 to seven field experiments on five working farms in the UK, for arable, legume, horticultural and root crops. **Results:** Three trials showed no significant ($p > 0.05$) effect on crop yield, two showed positive effects of 5–6%, one showed a very strong increase of 100% and one showed a decrease of 2–16%. A meta-analysis of effect sizes was conducted for all treatments ($n = 47$), which showed a significant ($p < 0.05$) positive effect, increasing average yield by 0.4 t ha^{-1}. Biochar application rates of 20 t ha^{-1} or under led to the greater benefits. **Conclusion:** This paper shows that, in some situations, biochar can bring benefits in modern temperate farming.

The **biochar** concept has attracted interest as a method for simultaneously decreasing atmospheric CO_2 levels at the gigaton scale [1,2] while increasing soil quality and crop productivity [3,4], producing bioenergy [5,6] and providing an efficient disposal route for wastes [7–9]. This paper focuses on crop responses to biochar, which may be a major motivation for land managers and farmers to use biochar unless some form of carbon payments are made. Much of the early research was performed in the South American tropics where the soils, climate and economics of farming are different to temperate regions of the world. Indeed, reviews [10,11] and meta-analyses of crop responses to biochar [12,13] have pointed out the lack of field trials in temperate regions.

In recent years there has been an increased interest on assessing physicochemical characteristics of different biochars [14–16], how they may be influenced by production conditions [17,18] and what effect this might have in the field [19–21]. Some studies addressed the mechanistic understandings [22] of how biochar interacts with various soil amendments [23,24], nutrient cycles [25–27], soil gas emissions [28–32], soil fauna [33–35] and plants [36], mostly at the laboratory scale. Other studies have estimated how much **pyrolysis** biochar systems might cost to deploy [37] and evaluated the potential benefits from the use of biochar by life cycle assessments [38,39]. While these studies have shown a favorable carbon balance – that is, net GHG reductions from the use of biochar – they have shown that the economics of applying biochar are not favorable in European countries. For land managers to pay to use biochar, or for the creation of sufficient subsidies, further work is needed on the understanding of biochar function in soil. Currently there is little published work on biochar field trials in temperate regions [40–42], and this study aims to contribute to this topic.

The present study focuses on crop yield response from seven field trials undertaken on five operational farms, during the period 2009–2012, with a total of 47 replicated biochar treatments (188 biochar plots in total). All the data gathered from each trial is not presented here in detail; only crop yields are reported at this stage. A meta-analysis of crop yields was also conducted to assess effect size of the results. The biochar used and the details of each trial are described in the 'Materials & methods' section, followed by the 'Results' and then 'Discussion' sections.

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Key terms

Biochar: Biomass that has been heated in a low-oxygen environment (pyrolysis or gasification), producing a charcoal-like substance intended for application to soils. Some groups have now produced formal definitions of biochar intended to certify certain products as 'biochar' so that consumers and regulators can be sure about what is, and what is not, biochar.

Pyrolysis: Thermal conversion (heating) of biomass in a low- or no-oxygen environment.

Cation exchange capacity: The maximum quantity of total of cations (e.g., NH_4^+, K^+ and Mg^+) that a soil is capable of holding, and which are available for exchange with the soil solution. More organic material in soils generally increases cation exchange capacity, but lower pH leads to decreases.

Materials & methods

Dalkeith charcoal fines

Biochar from the same production source was used on every trial. While this facilitates comparison between sites, it should be noted that different biochars may have led to different results. The biochar was produced by Dalkeith Charcoal (Dalkeith, Scotland), in charcoal-producing ring kilns between 2009 and 2011. The charcoal was destined for sale as barbecue fuel, but approximately one-fifth of the output was of particle size too small for sale; it was these 'charcoal fines' that were applied to the fields. The feedstock was mixed with deciduous and coniferous species including sycamore, oak, Scots pine and beech. The residence time for the char production was approximately 2 days and the peak temperature approximately 500°C. The process was partially pyrolytic and bioenergy co-products were not utilized. The char was stored in outdoor piles or in sacks in sheds until it was collected and transported to field sites. Dalkeith Charcoal ceased trading in 2011. Samples of Dalkeith char were assessed by the Scottish Environmental Technology Network Laboratory (Strathclyde University, Glasgow, UK) in April 2009 and again in September 2011 (Table 1). A representative 500 g sample of char underwent size fraction analysis to give an indication of the size proportions of char that were being handled and applied (Table 1). The sample tested in 2011 had been stored outside for an unknown period of time and may have changed due to oxidation, weathering or other factors during this period. The greatest change appears to be the **cation exchange capacity** and the exchangeable nutrient content of the biochar – each increased by one order of magnitude. This may be considered evidence for the beneficial 'aging' process of biochar. This particular sample was used in the Midlothian barley trial described below. It should be noted that the biochar was not tested prior to each field application, and that duration of storage varied for each treatment, although storage was generally in 20-l porous sacks inside barns. It is believed, but not proven, that the 2009 sample was representative of the biochar used in all trials except for the Midlothian barley trial.

Field trial summaries

Seven field trials were conducted in five locations in the UK, each is described in turn below. These trials were conducted under various separate research projects or programs and hence different variables were measured at different intervals in each trial. The comparability of much of the collected data is limited and therefore not discussed in this paper; results are discussed primarily in terms of crop yield response to biochar compared with the control, which was identical except that it received no biochar application. Crop yields are assessed trial by trial and all together through meta-analysis. Yield results from each trial were assessed for statistically significant difference with one-way ANOVA using the computer program Minitab 16. Where extensive data was collected from field trials it will be analyzed in greater detail and discussed in the meantime with the authors, to be published at a later date. Six of the seven trials received the full and usual application of fertilizers in the agricultural context according to normal farm practice; therefore, nutrient deficiency was not considered to be a limiting factor in those trials. The locations, farm conditions, trial objectives, methods and factors assessed are described below and summarized in Table 2.

East Lothian

The site in East Lothian, Scotland (latitude 56° 1'0.51"N, longitude 2° 40'40.50"W) is a working arable minimum-till farm using standard fertilizer additions [43]. The soil is classified as a silty clay loam. Experimental blocks were integrated into a 6.62-ha field and referenced using a differential GPS. The objective was to assess the economic efficacy of using biochar in a commercial agricultural setting. Accordingly, on-farm machinery was used for all operations and regular farming practices were maintained, except for the spreading of biochar onto some areas of the fields. Biochar was first deployed in March 2009 immediately prior to spring barley sowing (the biochar application rate was 10 t ha^{-1}). A further series of biochar plots were deployed in October 2009 (10, 20 and 40 t ha^{-1}) following the spring barley harvest and prior to sowing of winter oil seed rape. Both series of plots (10 × 8 m) were in the same field, had the same crops and amendments, and are otherwise treated the same. Plots were repeated in quadruple with a set of control plots that received no biochar. A disk broadcaster was used to apply the biochar, and biochar was integrated into the soil using a seed drill during the sowing operation. Prevention of dust during transportation and dispersal was reduced by addition of water to the biochar in the disk spreader. Correct quantity of biochar dispersed was achieved by adjusting broadcast settings on the machinery to the appropriate level for the density of the char. No further biochar has been added since 2009, but crop rotations have been continued and yields recorded (using a combine harvester with yield GPS tracking and yield

monitoring). The crops grown have been as follows: 2009 spring barley; 2010 winter oil seed rape; 2011 winter wheat. Soil pH, gravimetric moisture content and mineral nitrogen were assessed annually post-harvest.

East Lothian hectare trial

In September 2010, a 1-ha trial was initiated on an adjacent field to the already established plots. All crops and crop managements were the same as the other East Lothian trials in 2010 and 2011. Biochar was spread at 10 t ha^{-1} over half of the hectare, the other half used as a control plot (no biochar application), the objective being to test the effect of soil heterogeneity and edge effects on crop yield in comparison with the smaller plots. Application of the biochar was easier when using the same machinery as the East Lothian (smaller plot) trial, as the equipment could be utilized to its designed potential.

Angus

The farm in Angus (Scotland; latitude 56° 40'57.15"N, longitude 2° 37'7.24"W) is a commercial vegetable farm with a sandy loam soil. Biochar was applied at 0, 10, 20 and 40 t ha^{-1} in four replicates to beetroot (*Beta bulgaris*) and carrot (*Daucus carota* L.) crops in May 2009. The initial surface application of biochar was by hand for each plot (2 × 4 m, with 1 m space between plots). A bed former then incorporated the biochar during the formation of the beds and sowing. The bed former created a lot of drag and visual assessment showed that the biochar had been moved between plots. Biochar was still clearly visible in all plots, but it is uncertain that the plots can be considered representative of the intended application rates. As part of farm management, beetroot and carrot received 125 and 75 kg NH_4NO_3-N ha^{-1}, respectively. The major objective was to assess the effects of biochar on root vegetable crop yields, which were harvested by hand using a 0.25-m^2 quadrat. In addition, soil gravimetric moisture content, mineral nitrogen, pH, dissolved organic carbon and soil gases (N_2O, CH_4 and CO_2) were measured once per week.

Auchtermuchty

The farm in Auchtermuchty (Kingdom of Fife, Scotland; latitude 56° 29'06.15"N, longitude 3° 23'09.53"W) is an arable farm operating on standard principles [43], with a clay loam soil. Biochar was trialed during the 2010 season on spring barley crops at 0, 10, 20 and 40 t ha^{-1}, on 10 × 6 m plots, replicated in four blocks. The objective was to verify the outcomes seen on the East Lothian farm the previous year. Biochar was applied by hand and raked over the soil surface (March 2010), before incorporation by shallow ploughing followed by sowing and standard fertilizer application. Harvest was conducted by hand using a 0.25-m^2 quadrat, four per plot. The plant biomass was air dried and weighed. The barley heads were then removed and weighed separately.

Nottinghamshire

The Nottinghamshire field trial took place on a farm growing spinach (latitude 53° 7'43.13"N, longitude 1° 4'36.94"W) in a loamy sand soil. Spinach is a high-value crop and the farming system was markedly different to the others discussed in this paper. The spinach was grown during late April to early June 2010 over a 5-week period. The land was intensively managed, with macronutrients (nitrogen, phosphorus and potassium), micronutrients (e.g., molybdenum, magnesium, manganese and boron) and water (through large irrigation booms) delivered in precisely the quantities required for the crop. Before seeding, the soil was reformed into beds using a bed former towed by tractor, which grinds and reshapes the soil, homogenizing the soil structure. The growth process relies much more on inputs rather than inherent fertility of the soil. The objective was to assess how biochar might interact in such an already optimized system. Part of the motivation for the grower taking part in this experiment was his interest in reducing the carbon footprint of the crop.

Biochar was applied in two ways: deep incorporation and shallow incorporation. In both cases, the biochar was spread initially by hand on to the plots, which were 1.65 m × 5 m in size, with a 0.5 m separation buffer. Deep incorporation was to approximately 15 cm, and the biochar was incorporated by the bed former. Visual inspection suggests that the bed former dragged the biochar between plots. As with the Angus Farm trial, it is unlikely that the application rates specified accurately reflect the actual amounts in plots. Two additional control plots were selected, which

Table 1. Dalkeith charcoal analysis.

Date assessed	C (% dry weight)	H (% dry weight)	Ash content (% dry weight)	pH (electrochemical)	ICP-OES (cmol$^+$ kg^{-1})					Size fraction analysis (mm; % by mass of total)				
					Cation exchange capacity	Exchangeable				9.25 > 4	4 > 2	2 > 0.5	0.5 > 0.25	0.25 > 0.125	>0.125
						K	Na	Ca	Mg						
9 April 2009	85.75	3.22	5	9.29	11	3.4	0.4	4.7	2.5	16.5	28	38.7	10.6	3.1	3.4
5 March 2011	92.4	n/a	n/a	8.28	123.9	18.1	1.7	43.5	18.8						

ICP-OES: inductively coupled plasma optical emission spectrometry.

Table 2. Field trial summaries.

Trial location	Year	Crop	Soil type	Soil pH	Biochar application rate (t ha^{-1})	Biochar integration method	Biochar application depth (cm)	Date applied	Plot size (m × m)	Total biochar area (m^2)	Management	Trial objective
East Lothian	2009	Spring barley	Silty clay loam	7.39	10	Seed drill	8	March 2009	10 × 8	320	Mininimum-till arable	Biochar effect on working farm economics
	2010	Oil seed rape		7.39	10, 20 , 40	Seed drill	8	October 2009				
	2011	Winter wheat		7.39	10, 20 , 40	Seed drill	8					
East Lothian: hectare	2011	Winter wheat	Silty clay loam	7.39	10	Seed drill	8	September 2010	100 × 50	5000	Mininimum-till arable	Biochar effect at field scale
Angus	2009	Carrot	Sandy loam	6.25	10, 20, 40	Bed former	45	May 2009	2 × 4	288	Standard	Effect on root vegetables
		Beetroot		6.25								
Auchtermuchty	2010	Spring barley	Clay loam, some peat		10, 20, 40	Shallow plough	25	March 2010	10 × 6	720	Standard arable	Effect on cereal
Nottinghamshire	2010	Spinach (shallow)	Loamy sand	7.2	6, 12, 24, 40	Seed drill	3	April 2010	1.65 × 5	264	Intensive	Biochar effect on high-value crop
		Spinach (deep)		7.2	12, 24, 40, 80	Bed former	15					
Midlothian: legume	2011	Winter beans	Loam	6.14	10, 30	Seed drill	8	November 2010: winter crops	2 × 3	432	Green manure	N-fixation and retention from legumes
		Spring beans, vining peas, combining peas		6.14	10, 30	Seed drill	8	March 2011: spring crops				
Midlothian: barley	2011	Spring barley	Loam	6.06	10, 30	Top-dressing	0	March 2011	3 × 4	360	High and low NH_4NO_3 applications	Efficient N usage from mineral fertilizer

were a distance of 5 m beyond the final plots along the row, to minimize any interference from dragged biochar. For shallow incorporation the biochar was spread on the surface of the formed beds and incorporated through seed drilling (to approximately 3 cm soil depth). The biochar application rates for deep incorporation were 0, 12, 24, 40 and 80 t ha^{-1}. At shallow incorporation the biochar application rates were lower at 0, 6, 12, 24 and 40 t ha^{-1}. It was considered that shallow incorporation of biochar at 80 t ha^{-1} would have inhibited germination. A lower rate of biochar application was therefore tested.

Each plot was repeated in quadruple and placed randomly along two adjacent rows. A bespoke spinach harvester was used to harvest the spinach mechanically. As the crop from each plot was cut, it was weighed in-field.

Midlothian: barley trial

A spring barley (*Hordeum vulgare*) trial was established in the Midlothian region of Scotland, southwest of Edinburgh (latitude 55° 52'12.61"N, longitude 3° 12'13.25"W) during 2011. The Midlothian site (loam soil) is an experimental farm operated by Scotland's Rural College, which unless otherwise requested is run on standard agricultural practice. Biochar was manually applied to the surface of the field plots (3 × 4 m) at two rates of 10 and 30 t ha^{-1}, and was applied in conjunction with five nitrogen fertilization (NH_4NO_3; 34.5% nitrogen) treatments (repeated in triplicate), which were relative in percentage terms to the recommended fertilization rate (0, 60, 120, 180 and 240% of the standard 120 kg N ha^{-1} [43]) for spring barley in Scotland.

The biochar application was a top-dressing – it was not incorporated into the soil by any mechanical means, minimizing the risk of drag and migration of biochar between plots. While surface application is an interesting option for reasons discussed below, it can only last for one season as the soil will be turned before the next seeding, and the biochar is potentially at risk of being washed or blown away, although this was not observed during the trial.

A 0.125-m^2 quadrat was used to measure crop yields, one taken per plot. The quadrats were placed randomly within plots and all biomass within the quadrat was harvested, separated into constituent parts (straw, grain and chaff), dried at 65°C for 48 h and weighed. Soil gravimetric moisture content, mineral nitrogen and gases were measured on a frequent basis (frequency depending on the season; full results remain to be published).

Midlothian: legume trial

A biochar-legume trial was also established at the Midlothian farm (latitude 55° 52'12.61"N, longitude 3° 12'13.25"W). Winter beans (*Vicia faba* cv. Wizard) were sown in November 2010, and spring legumes – combinable peas (*Pisum sativum* cv. Crackerjack), vining peas (*Pisum sativum* cv. Oasis) and spring beans (*Vicia faba* cv. Fuego) were sown in March 2011. Biochar was applied by hand at 10 and 30 t ha^{-1} to the surface of each plot (2 × 3 m) and incorporated into the soil using a seed drill. Bean and pea yields were taken using a 0.25-m^2 quadrat, one per plot, with the crop separated into constituent parts (straw, seed and pods), dried at 65°C for 48 h and weighed. Additionally, soil gravimetric moisture content, mineral nitrogen and gases were measured on a frequent basis (frequency depending on the season; full results remain to be published). The main objective of this experiment was to assess the effect of biochar on yields and nitrogen fixation by legumes, and subsequent nitrogen losses (or retention) following the year of growth. The biochar plots were 2 × 3 m and repeated in triplicate. Following harvest the soils were ploughed, which caused considerable biochar movement, mainly biochar moving approximately 3 m laterally from the biochar plots into the nonbiochar plots, with a higher concentration of biochar closer to the biochar plots. This, however, does not affect the results presented in this paper, but the following season's results.

▪ Meta-analysis of effect sizes

Meta-analysis is used as a tool for the pooling of information from independent studies. Although usually used for literature reviews, the seven field trials reported here were conducted as independent studies and may be analyzed succinctly through meta-analysis. For this meta-analysis the control was considered identical to the experimental treatments in relation to all variables except the addition of biochar.

The commercial Comprehensive Meta-Analysis software program was used to assess the dataset. To perform a meta-analysis, the effect size and variance for each study is computed followed by a computation of a weighted mean of the effect sizes. The effect size (the standardized difference in means, in this case measured in t ha^{-1}) is a value that reflects the magnitude of a treatment effect against a control treatment, dispersion of effects across studies or (more generally) the strength of a relationship between two variables [44].

Results

▪ Field trials

The crop yield results for each trial are presented in Table 3. In most cases, the crop yield response to biochar addition was not statistically significant, showing in most cases no effect on crop yield due to the applied biochar compared with the control. Statistically significant ($p < 0.05$) effects were found at East Lothian in 2011, the Midlothian barley trial (2011) and in the Nottinghamshire spinach trial (2010).

The East Lothian 2011 crop of winter wheat showed a 1–6% yield increase with biochar additions of up to 20 t ha^{-1}. The hectare trial at East Lothian also showed a yield increase of 4.6% for biochar addition of 10 t ha^{-1} that year. The yield increase with 10 t ha^{-1} of biochar was larger than the yield increase with 20 t ha^{-1}. Interestingly, biochar applications of 40 t ha^{-1} resulted in a yield decrease of 3%. It is also worth noting that the same site did not show significant yield responses in the previous 2 years.

The Midlothian barley trial showed the strongest effects of biochar applications of any trial reported in this paper, with the crop yield more than doubling. Biochar applications of 10 t ha^{-1} showed statistically significant positive effects on crop yield across all five rates of nitrogen fertilization and biochar application of 30 t ha^{-1} showed statistically significant positive effect only at the three higher rates of fertilizer addition. Average yield increases were 117% for 10 t ha^{-1} of biochar and 107% for 30 t ha^{-1} biochar additions. Both biochar treatments maintained soil moisture at a level consistently higher than the control plots throughout the duration of the experiment (full results remain to be published).

Table 3. Crop yield results and one-way ANOVA significance values.

Trial location	Year	Crop	Biochar application rate (t ha^{-1})	Mean yield (t ha^{-1})	% change to control yield	p-value	Number of replicates
East Lothian	2009	Spring barley	0	7.88		0.287	4
			10	8.43	6.96		8
	2010	Oil seed rape	0	3.39		0.427	4
			10	3.71	9.31		8
			0	3.36		0.583	4
			10	4.12	22.62		4
			20	3.72	10.75		4
			40	3.52	4.82		4
	2011	Winter wheat	0	8.81		0.008	4
			10	9.28	5.28		8
			0	8.78		0.001	4
			10	9.31	6.06		4
			20	8.88	1.14		4
			40	8.51	-3.03		4
Angus	2009	Carrot	0	16.55		0.644	4
			10	18.8	13.6		4
			20	17.2	3.93		4
			40	18.33	10.73		4
		Beetroot	0	8.86		0.268	4
			10	8.64	-2.46		4
			20	10.35	16.87		4
			40	8.8	-0.63		4
Auchtermuchty	2010	Spring barley	0	8.87		0.954	16
			10	8.9	0.28		16
			20	9.09	2.45		16
			40	8.87	-0.03		16
Nottinghamshire	2010	Spinach (shallow)	0	30.61		0.960	5
			6	29.96	-2.11		4
			12	29.23	-4.52		4
			24	29.38	-4.03		4
			40	29.34	-4.15		4
		Spinach (deep)	0	31.74		0.022	5
			12	26.51	-16.48		4
			24	30.71	-3.25		4
			40	28.44	-10.41		4
			80	30.86	-2.78		4
East Lothian: hectare	2011	Winter wheat	0	8.316		n/a	1
			10	8.696	4.57		1

Table 3. Crop yield results and one-way ANOVA significance values (cont).

Trial location	Year	Crop	Biochar application rate (t ha^{-1})	Mean yield (t ha^{-1})	% change to control yield	p-value	Number of replicates
Midlothian: legume	2011	Winter beans	0	5.31		0.068	4
			10	7.51	41.33		8
			30	5.41	1.9		8
		Spring beans	0	5.38		0.903	4
			10	5.28	-1.86		8
			30	4.65	-13.45		8
		Vining peas	0	1.79		0.106	4
			10	1.41	-21.32		4
			30	1.35	-24.55		4
		Combining peas	0	7.28		0.746	4
			10	7.36	1.13		4
			30	8.76	20.24		4
Midlothian: barley	2011	Spring barley	0	4.429		0.000	15
			10	9.595	116.64		15
			30	9.111	105.71		15

The Nottinghamshire spinach farm showed a statistically significant decrease in crop yields when biochar was incorporated at 15 cm depth. There was no clear pattern relating biochar application rate to lower yield, which ranged from 2.8 to 16.5%. Although not statistically significant, shallow biochar incorporation also showed lower yield for all treatments (Table 3).

Results of meta-analysis

Meta-analysis of effect sizes for all the field treatments across the trials are shown in Figure 1. The analysis is made up from 47 different treatments and 188 replicates in total. Each point represents the effect size of biochar-treated plots versus control plots in relation to yield (t ha^{-1}), with the lines representing 95% confidence intervals. Where confidence intervals cross 0.00 the effect is not significant. The overall effect for all treatments is marked by a red diamond. Out of the 47 treatments reported here, nine resulted in significant crop yield increases ($p < 0.05$), three showed significant decreases and 35 showed no significant effect. The overall effect size shows a significant ($p < 0.05$) effect of biochar treatment versus control plots upon the resultant yield of 0.4 t ha^{-1}.

Discussion

From these findings, it appears that multiple interacting factors are altered by addition of biochar and that some of these factors constrain crop yield or growth in certain situations. The efficacy or otherwise of biochar can rarely be put down to a single factor. Those factors are soil texture, soil pH, nutrient balance, micronutrient balance, weather conditions, biochar aging, biochar application rate, biochar incorporation depth and crop type, as well as type of biochar and other unknown factors. Due to a limited dataset and dominance of the Midlothian barley trial, it was not possible to determine the statistical influence of each individual factor in relation to the impact of biochar application upon yield in the present paper (as was performed, e.g., in Jeffery *et al.* [12]).

Most of the cases reported here resulted in no significant effect on crop yield from biochar addition. If no improvement on this trend is achieved then the purpose of using biochar in temperate regions will be primarily for carbon sequestration, and in some cases compaction and disposal of wastes. Although the outcome of 'no significant effect' on crop yield is the common finding for biochar trials in the temperate regions [40], some of the trials reported here did show significant positive effects on crop yield, implying that it is possible to achieve under certain circumstances. A better understanding of what those circumstances may be would be very useful. While few conclusions or firm statements can be drawn from the work presented here, some observations can be made which, it is hoped, might prove useful for future research.

The meta-analysis of effect sizes for the field data showed that most trials demonstrated no significant effect, but when compiled showed an overall positive effect size of 0.4 t ha^{-1} for biochar treatment upon crop yield ($p = 0.043$). This effect size is smaller than the average 10% increase reported in the meta-analysis of biochar trials in tropical regions [12]. The finding from the present study has been strongly influenced by the large effect seen in the Midlothian barley trial. The dominant finding of this study is that biochar commonly shows no effect on crop yield in the studied conditions, but that it can increase crop yield in some circumstances, and can increase it considerably.

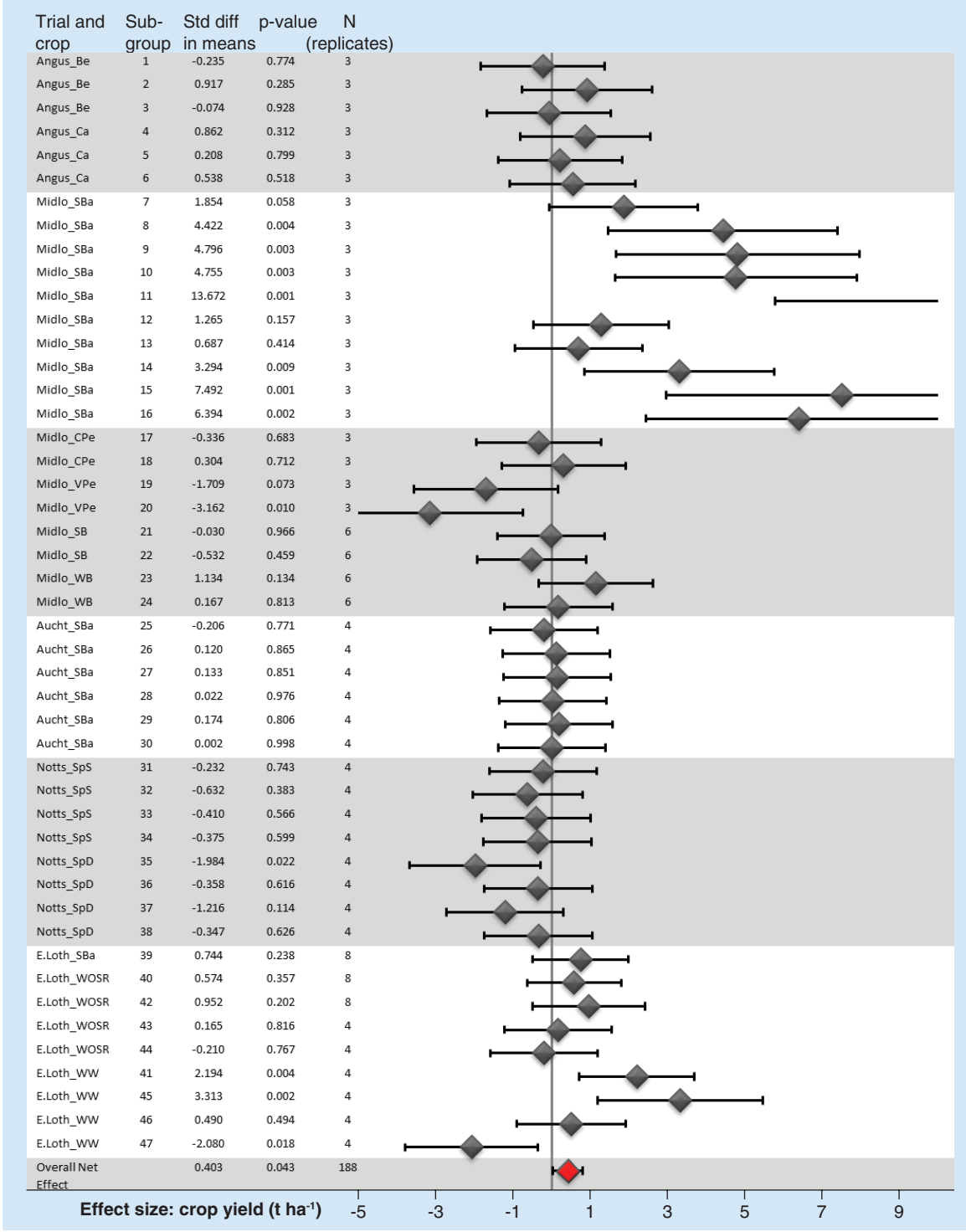


Figure 1. All trials (n = 47) showing the effect size of crop yield (t ha^{-1}) for biochar treated plots versus control plot yield. Each point represents the effect size with the lines representing 95% confidence intervals. If the confidence interval crossed 0.00 t ha^{-1} then the effect is not significant. The overall effect is marked by a red diamond.
The study name indicates the location and the crop grown.
Aucht: Auchtermuchty; Be: Beetroot; Ca: Carrot; CPe: Combinable pea; E.Loth: East Lothian; Midlo: Midlothian; Notts: Nottinghamshire; SB: Spring bean; SBa: Spring barley; SpD: Spinach (deep); SpS: Spinach (shallow); Std diff: Standard difference; VPe: Vining pea; WB: Winter bean; WOSR: Winter oilseed rape; WW: Winter wheat.

The most striking field result is that of the Midlothian barley trial, although it should be noted that both control plot yields and biochar plot yields were within the range of 'normal' for that crop and region [43]. Two things were unique about this trial: the biochar was stored outside for a period of years and showed an order of magnitude higher cation exchange capacity and exchangeable nutrients (Table 1); and the biochar was top-dressed for soil application. Top-dressing of the biochar may have had a number of effects such as: fertilizers were applied to the soil surface after the biochar so some reactions with the biochar may have occurred; the layer of biochar may have increased soil temperature, especially early in the season, leading to higher rates of germination and successful early establishment; and the biochar layer may have reduced evaporation from the soils, perhaps increasing soil moisture. Biochar has a recorded effect on soil moisture, generally increasing the water-holding capacity of soils [7,45]. However, drawing on local weather station data and comparing with a 30-year average, it can be seen that the rainfall for March–September 2011 was 200% of the usual rainfall, so while lack of water is not likely to have been a constraint, perhaps some effect of the biochar was beneficial during wet periods too. The soil texture was loamy and the soil pH was 6.06. Medium textured soils showed highest response to biochar additions [12], and the soil pH would have been raised by addition of the basic biochar. Biochar commonly shows some benefit in soils with lower pH [12,46].

While the Midlothian legume trial was conducted at the same site during the same season, there was no significant effect of biochar upon yield response. The biochar used had a lower cation exchange capacity, was incorporated to 8 cm, and legumes were grown instead of barley. Legumes have previously been shown to respond positively to biochar addition [47].

The East Lothian trial showed significant yield effects in 2011, but not in 2010 or 2009. One conclusion to draw could be that certain crops respond better to biochar than others; however, this would be an oversimplification. Spring barley was grown in East Lothian 2009 and Auchtermuchty 2010 and showed no significant effect to biochar, but when grown in Midlothian in 2011 showed the greatest response reported in this study. Indeed, East Lothian and Midlothian both showed positive yield effects during the season of 2011. The two sites are within 45 km and so it is possible that the same weather patterns affected both sites. Perhaps the effect biochar can have on water-holding capacity of soils influenced crop growth during that season [7,48]. An alternative hypothesis is related to biochar 'aging': a process whereby biochar increases in cation exchange capacity and becomes 'charged' with nutrients [48–50]. It may be that by 2011 the biochar at East Lothian farm had aged sufficiently to give beneficial effects by the third year in the soil and that the biochar applied to Midlothian had aged sufficiently during storage (as evidenced by the analysis in Table 1) to deliver crop benefits.

Key term

Nutrient immobilization: Usually refers to the conversion of nutrients from an inorganic, plant-available form into an organic form (i.e., by microbial of fungi communities) unavailable to plants. There may be some 'biochar immobilization' effect during the 'aging' process during which nutrients are made unavailable to plants.

Some negative effects were seen from higher application rates of biochar. This is most clear in the East Lothian 2011 trial, where application rates of 10 and 20 t ha^{-1} resulted in yield increases but application rates of 40 t ha^{-1} resulted in a yield decrease. It is also seen in East Lothian 2010, the Midlothian barley trial and Auchtermuchty. It may be that at higher rate of biochar addition **nutrient immobilization** occurs [36,49,51]. This is contrary to the findings of Jeffery *et al.,* who showed increasing benefit with increasing application rate [12].

The Nottinghamshire spinach farm showed lower yield due to biochar. The farming operation relied on the tightly specified and controlled delivery of nutrients, micronutrients and water at prescribed times as required. It is feasible that the biochar may have disrupted this balance. These results therefore suggest that the intensively managed spinach farming system was not appropriate for biochar addition. Although the soil texture was sandy, which typically responds well to biochar addition [12,52], the biochar in this case was not beneficial, possibly because the 'constraints' of the sandy soil (poor nutrient content and water-holding capacity) had been overcome using a different method (frequent and optimized application of nutrients and water).

Conclusion: implications for use of biochar

The same biochar was tested in a range of different crops and soil types, on five farms and in seven separate experiments over a 3-year period. Three trials showed statistically significant positive crop yield responses, one trial showed significant negative responses and the other trials showed no significant response. Although no effect on crop yield is the most common finding here, the results show that both positive and negative effects on crop yield are achievable in the temperate region. Using the extra statistical leverage of a meta-analysis, an overall positive and statistically significant ($p < 0.05$) effect of biochar on crop yield of 0.4 t ha^{-1} was found, although this may have been due to the strongly positive outcome of one trial. A better understanding of why biochar causes effects in some situations and not others would be very useful.

Based on the experience of conducting biochar field trials, practical advice is offered in Box 1. Based on the results observed we offer the following implications for the use of biochar, which we consider to be based on good evidence:

Box 1. Practical issues to consider in biochar field trials.

- Biochar is highly porous and can absorb many times its own weight in water. One ton of 'dry' biochar could weigh ≥2 tons wet. Volumetric measurements of biochar are therefore more reliable as moisture content may change with storage.
- It is desirable for biochar to be moist during storage as it decreases the fire risk; and moist biochar during field application reduces the risk of loss from wind erosion and lowers the dust content.
- When applying biochar it is useful to use sacks of known volume. One sack of dried biochar can be weighed (or an average taken of many), and then these sacks can be used to spread a known 'dry mass' of biochar onto a plot irrespective of moisture content (since the volume of biochar does not increase with increasing moisture content).
- Displacement of biochar around field plots can occur when using machinery, such as ploughs or bed formers, which move soil around the field. Displacement occurs in the direction of the tractor travel and laterally to the direction of tractor travel.
- Plots of 10 × 10 m are recommended, with at least 2 m buffer strips between. Plots under 5 × 5 m or without buffers are not recommended.

- Biochar application at 10 t ha^{-1} can be sufficient to produce benefits, or at least no negative effects. Beyond 30 t ha^{-1} can produce negative effects;
- Shallow incorporation and top-dressing can be sufficient to produce benefits. Deeper incorporation is not necessarily required to achieve benefits;
- Multiyear trials yield more information and are needed to understand both the effect of biochar aging in soils and the prevailing weather conditions in different years;
- Crops in very intensively managed systems (crops that already have everything they need) are unlikely to benefit from biochar application;
- Machinery such as ploughs and bed formers can move biochar meters from its original position. Where possible, use large plots (e.g., 10 × 10 m) with buffer strips between, or use shallow incorporation methods;
- Measuring biochar by mass is deceptive; the water content can be double or more of the biochar mass with no increase in volume. Volumetric measurements are more reliable.

The following we offer tentatively:

- Arable crops seem to respond better than root or horticultural crops, although this may be an artefact of the way the crops are managed and of the fewer trials with non-arable crops;
- Biochar may provide a buffering function in soils during periods of low or heavy rainfall;
- Top-dressing of biochar is worthy of further investigation;
- Biochar will be more effectively deployed if it is used to meet some need from the crop – that is, if it used to remove some constraint to increased crop growth.

Future perspective

During the past few years the published work on biochar has grown exponentially. This trend is expected to continue, and understanding of how biochar functions in soil will continue to improve. Further field trials are under way at present and findings will be published in the years to come. The use of biochar in combination with other amendments, especially organic amendments such as compost, slurry or manure, but also incorporation into fertilizer pellets, will increase in sophistication and is expected to become the main route for biochar to enter soils. Production of biochar from waste materials is expected to increase, especially combinations of nutrient-rich material such as sewage sludge or chicken litter with lignocellulosic material. As the use of biochar becomes more nuanced, site selection will become more targeted and beneficial results will hopefully be achieved more consistently. Biochar may play a greater role in the organic farming sector where greater emphasis is placed on soil humus levels, microbes and nutrient cycling. If subsidy payments are made for the stable carbon in biochar then use will increase at a much greater rate than if farmers have to pay for biochar production and application.

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Executive summary

Materials & methods

- One type of wood chip biochar was applied to field sites on five working UK farms for a range of different crops soil types during 2009–2011.
- Seven experiments were conducted, totalling 47 different biochar/crop replicated treatments and 188 biochar plots in total.

Results

- Of the seven experiments, three showed no significant effect of biochar on crop yield, three showed significant improvements in crop yield and one showed a significant decrease in crop yield.
- Meta-analysis of all treated field plots showed an overall significant positive effect on crop yield of 0.4 t ha^{-1}.

Discussion

- Biochar application rates below 30 t ha^{-1} showed more improvement to crop yield than higher rates. Surface and shallow (<8 cm depth) incorporation of biochar was sufficient to show beneficial effects; deeper incorporation did not appear necessary.
- Crop yield responses varied from 100% increase to 24% decrease, indicating that to gain the greatest benefit from biochar it must be used in the correct context.
- The most intensively managed crop responded the worst to biochar; biochar may have disrupted the optimzed nutrient and water balance.
- An understanding of the constraints to higher yields and the role of biochar in alleviating those constraints would be useful.

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